IMPLEMENTING PID POLICY IN DATAVERSE: CHALLENGES AND OPPORTUNITIES

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Introduction

National and international requirements and recommendations for PID issuing:

- **FAIR (F1):** (Meta)data are assigned a globally unique and persistent identifier
- A Persistent Identifier (PID) policy for the European Open Science Cloud **(EOSC)**
- **CESSDA ERIC** Persistent Identifier Policy and Best Practice Guidelines for Service providers

Organizations that employ **Dataverse software** for data curation in their repositories are dependent of the software when formulating their PID issuing policy:

- DOIs and Handles only supported in Dataverse
- Data sets receive PIDs by default
- Files with PIDs - optional

Problem

How to develop PID policy for a repository based on Dataverse software compliant with the requirements and recommendations for best practise?

Exercise

FAIR-IMPACT project Support Action: Creating EOSC compliant Persistent Identifier (PID) policies

Aim - to create PID Policy for Data Sets in the Repository of Lithuanian Data Archive for Social Sciences and Humanities (LiDA, https://lida.dataverse.lt), which is based on Dataverse software

- PID policy should comply to EOSC PID Policy and CESSDA ERIC PID Policy
- PID policy should take into consideration opportunities and restrictions of the Dataverse software

The Dataverse Project cessda eosc

Results of Exercise

- CESSDA ERIC PID Policy, Best Practice Guidelines and Checklist for the Usage are very handy documents for a start

> Section 1: Basics
>
> **Do you know what kind of PID system is right for you?**
>
> PIDs refer to a range of identifier systems. Common ones are Digital Object identifier (DOI) and ePIC, both based on the Handle system. Others include Universal Resource Name
>
> (URN), and Archival Resource Key (ARK). However, all PID systems require management and maintenance and cannot be left to run automatically.
>
> DOIs may be the most popular form of PID for data archives, and the Handle system on which it is based is a runner up, but that does not mean it is the right choice for your organisation - or even the exclusive choice.
>
> Choice is also based on the type of objects you have, what you want the PID to do, consideration of your technical infrastructure, level of financial commitment, value to your user community, and archival value.

- EOSC PID Policy is a more abstract document encompasing not just PID managing institutions
- However, its Guidelines for Implementation of Criteria (12 of the 35) for PID Managers (CAT) are relevant and handy

#	Criterion	Description	Imperative
C5	Update Functionality	The PID Manager **MUS**T provide the functionality required to maintain PID attributes. (Also mapped to the Actor: Service)	MUST
C6	Ownership Transfer	The PID Manager **SHOULD** provide policies and contractual arrangements for transfer of ownership should the owner no longer be able to assume responsibilities in compliance with the policy.	SHOULD
C7	Resolution Integrity	The PID Manager **MUST** maintain the integrity of the relationship between entities and their PIDs, in conformance to a PID Scheme defined by a PID Authority.	MUST
C11	Versioning - Procedure	PID Services and PID Managers **SHOULD** have clear versioning policies. (Also mapped to the Actor: Service)	SHOULD
C14	Resolution Authenticity	PID Manager **MUST** ensure that the entity remains linked to the PID. In case that the entity being identified is deleted or ceases to exist, tombstone information needs to be included in the PID attribute set.	MUST
C16	Digital Representation	Physical and conceptual entities **MUST** be represented via a digital representation (e.g. landing page, metadata, attribute set, database index) to have a presence in the digital landscape.	MUST
C19	Accurate Entity Metadata	The PID Service **MUS**T maintain entity metadata as accurately as possible in collaboration with the PID Owner. This copy is the authoritative version.	MUST
C22	No End User Cost	The basic services of PID registration and resolution **SHALL** have no cost to end users.	SHALL
C28	Certification	PID Authorities and Services **MUST** agree to be certified with a mutually agreed frequency in respect of policy compliance. (Also mapped to the Actors: Scheme, Authority, Service, Owner)	MUST
C29	Agreed Responsibilities	PID Services **SHOULD** agree with PID Managers the responsibilities for Kernel Information maintenance, preferably via contract (Also mapped to the Actor: Service)	SHOULD
C34	Persistent Median	PID Services **SHOULD** aim for a persistence median that is acceptable to and aligns with community and dependency	SHOULD

In its current version (6.4) **Dataverse software** allows issuing globally unique persistent identifiers for data sets and data/documentation files within datasets

- PID issuing implementation **for data sets is compliant** with requirements and recommendations of best practise
- PID issuing **for files within data sets is not compliant** with requirements and recommendations of best practise (for example, no tombstone page, little metadata)
- PID issuing **for collections of data sets is not implemented**

LiDA Persistent Identifier Policy for Data Sets developed: https://zenodo.org/records/13827904

- In addition, **action plan** for monitoring and implementing required changes or improvements published (and constantly updated) alongside the policy
- 4 types of actions identified: No action needed, Attention needed (every two years); Care needed (certain workflows for publishing); Immediate action needed (in order for practice to align with the policy)

Solutions needed in Dataverse

- **PID issuing for files within data sets should be improved**
- **PID issuing for collections of data sets should be implemented**

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Support

eosc | FAIR-IMPACT Expanding FAIR solutions across EOSC

FAIR-IMPACT project Support Action: Creating EOSC compliant Persistent Identifier (PID) policies (https://fair-impact.eu/support-offer-2-creating-eosc-compliant-persistent-identifier-pid-policies).

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